

**Gurdwara Gur Nanak
Prakash Tracy
Robotics Team**

About Me

- Lead Field Technician for Quarry Lane High School Robotics Team
- Part of the Design & Electrical Division
- Conducted research on tactile training for quadrupeds with Professor Cao from Case Western Reserve University
- Alameda County Science Fair
 - Humanoid Robot to Fight Fires → 2nd Place
 - Tree Planting Robot to Combat Climate Change → 1st Place



What is FIRST Robotics

Why?

- Very little Sikh representation at robotics tournaments
- Robotics is the future of this world with companies such as
 - Figure
- As a community we have to be part of this push in technology
- Good way for kids to get into Engineering or STEM related fields



**Start off with
creating Small
Robots**

**Advance through
Regionals**

**Compete at 2026
FTC World
Championships**

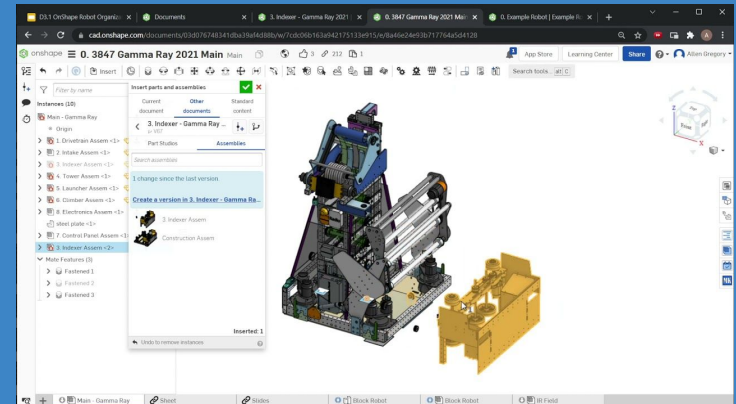
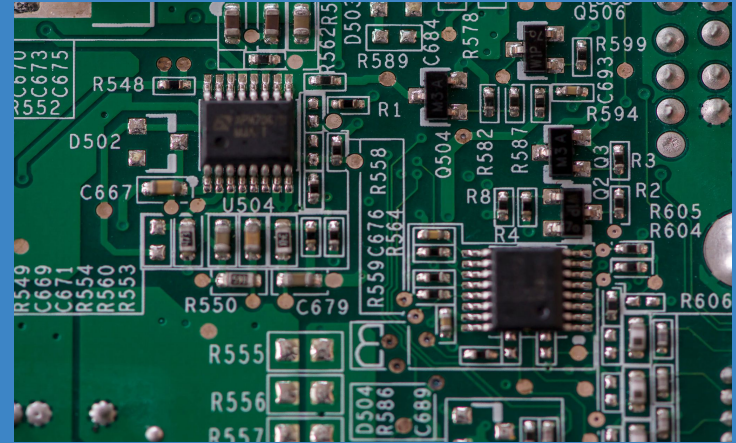
**Create a
Competitive
Robotics Team**

**Advance through
Districts**



What Will Be Taught

- Basic electrical concepts/techniques
- Programming specific to robotics, starting off with Python
- Work with Opencv, vision and autonomous navigation
- Mechanical Design Principles in Onshape
- Learn how to use shop machines such as a drill press and bandsaw



Next Steps

- Run Robotics Classes as part of Khalsa School
- Create a competitive FIRST Robotics Program here to compete at World Championships
- Team will require people interested in Business, Finance, Engineering, Programming, etc
- Setup our own makerspace for students to come and work on their own personal projects and work on the robot for the team
- If you are interested please sign up on the sign up sheet